



WSQ Courseware Generator

Complete User Guide

Step-by-Step Instructions for All Features

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1. Getting Started

1.1 Opening the App

The WSQ Courseware Generator runs as a web application in your browser. To start the app:

Step 1: Open a terminal / command prompt in the project folder.

Step 2: Run the command: `uv run streamlit run app.py`

Step 3: Your browser will open automatically at **http://localhost:7860**

***Note:** If the browser does not open automatically, manually go to `http://localhost:7860` in Google Chrome.*

1.2 Selecting a Company

At the top of the left sidebar, you will see a **Company dropdown**. This determines which company's branding (logo, UEN, address) appears on all generated documents.

- Select the correct company **before** generating any documents.
- The default is 'Tertiary Infotech Academy Pte Ltd'.
- To add or edit companies, see **Section 10: Company Management**.

1.3 Understanding the Sidebar

The left sidebar contains the main navigation menu with 8 pages. Below the menu, you will find:

- **Companies button** — Opens company management settings.
- **Agent Status** — Shows if any AI background job is running.

***Note:** The recommended workflow follows the menu order from top to bottom: Extract Course Info → Generate AP/FG/LG → Lesson Plan → Assessment → Slides → Brochure → Audit.*

2. Extract Course Info

This is **always the first step**. You must extract course information from an approved Course Proposal (CP) before generating any courseware documents. The extracted data is automatically shared across all other pages.

2.1 Uploading a Course Proposal

Step 1: Navigate to **Extract Course Info** in the sidebar.

Step 2: Click the **file upload area** (drag-and-drop also works).

Step 3: Upload your Course Proposal file. Supported formats:

- **.docx** — Word document (most common)
- **.xlsx** — Excel spreadsheet

2.2 Entering Course Details

Step 1: Enter the **Course Reference Code (TGS)** — e.g., TGS-2024001234. This code appears on all generated documents.

Step 2: (Optional) Enter the **Course URL** — the link to the course webpage. This is only needed if you plan to generate a brochure later.

Step 3: Click the **Extract** button to start processing.

The system uses AI (Claude Agent) to interpret the Course Proposal and extract structured data. This runs in the background — you will see a **spinning indicator** while it processes. You can wait on this page or navigate away; the extraction will continue.

2.3 Reviewing Extracted Data

Once extraction completes, you will see a **success card** showing:

- Course Title
- Number of Learning Units (LUs)
- Number of Topics
- Assessment Methods

Below the success card, expandable tables show the full extracted data: course overview, learning outcomes, topics with bullet points, and assessment details. **Review this data carefully** — it forms the basis for all generated documents.

Tip: If the company name in the CP matches a company in the database, the sidebar company dropdown will auto-select it for you.

3. Generate AP / FG / LG

This page generates three core WSQ courseware documents using AI agents:

- **Assessment Plan (AP)** — Outlines assessment strategy, methods, and criteria.
- **Assessment Summary Record (ASR)** — Auto-generated alongside the AP.
- **Facilitator Guide (FG)** — Instructor manual with lesson content and activities.
- **Learner Guide (LG)** — Student handbook with learning content.

3.1 Selecting Documents to Generate

Step 1: Navigate to **Generate AP/FG/LG** in the sidebar.

Step 2: You will see the extracted course info summary at the top. If it says 'No course info loaded', go back to Step 2 first.

Step 3: Use the **checkboxes** to select which documents to generate:

- Learning Guide (LG) — checked by default
- Assessment Plan (AP) — checked by default
- Facilitator Guide (FG) — checked by default

***Note:** You can uncheck any document you don't need. For example, if you only need the LG.*

3.2 Running Generation

Step 1: Click the **Generate Documents** button.

Step 2: A spinner will appear as each document is being generated by the AI agent.

Step 3: Generation takes 1-3 minutes per document depending on course complexity.

The system automatically enriches the data with TSC (Training & Skills Competency) fields, proficiency levels, and sector information before generating.

3.3 Downloading Documents

Once generation completes, you have two download options:

- **Individual downloads** — Click the download button next to each document.
- **Bulk download** — Click **Download All as ZIP** to get all documents in one file.

Files are named as follows:

- LG_[TGSCode]_[CourseName].docx
- Assessment_Plan_[TGSCode]_[CourseName].docx
- Assessment_Summary_Record_[TGSCode]_[CourseName].docx
- FG_[TGSCode]_[CourseName].docx

Tip: All generated documents are also automatically saved to the Courseware/ folder in the project directory.

4. Generate Lesson Plan

The Lesson Plan is generated **instantly** using a pure Python scheduling algorithm — no AI processing is needed. It creates a day-by-day timetable for the course.

4.1 Generating the Schedule

Step 1: Navigate to **Generate Lesson Plan** in the sidebar.

Step 2: The extracted course info is auto-loaded. Verify the course details at the top.

Step 3: Click **Generate Lesson Plan**.

The schedule follows these rules:

- Daily hours: **9:00 AM – 6:00 PM**
- Lunch break: **12:30 PM – 1:15 PM** (45 minutes, fixed)
- Assessment: **4:00 PM – 6:00 PM** on the last day only
- Topics are distributed evenly across available time
- Breaks fill all remaining gaps to ensure a full 9am–6pm day

4.2 Previewing & Downloading

After generation, you will see:

- **Statistics** — Total days, minutes per topic, instructional hours, assessment hours.
- **Day-by-day preview** — Expandable sections showing each day's timetable.
- **Download button** — Downloads as LP_[TGSCode]_[CourseName]_v1.docx

5. Generate Assessment

Generate assessment question papers in standard WSQ format. Supports 9 assessment types.

Code	Assessment Type	Description
WA (SAQ)	Written Assessment	Short answer questions testing knowledge
PP	Practical Performance	Hands-on task demonstration
CS	Case Study	Scenario-based analysis questions
OQ	Oral Questioning	Verbal Q&A assessment
OI	Oral Interview	Structured interview assessment
DEM	Demonstration	Live skill demonstration
RP	Role Play	Simulated scenario performance
PRJ	Project	Extended project-based assessment
ASGN	Assignment	Written assignment submission

5.1 Uploading Facilitator Guide (Optional)

You can optionally upload the Facilitator Guide (FG) to extract Knowledge (K) and Ability (A) statements. This helps the AI map assessment questions to specific K/A items.

Step 1: Navigate to **Generate Assessment** in the sidebar.

Step 2: (Optional) Upload the FG .docx file to extract K/A statements.

5.2 Selecting Assessment Types

Step 1: Use the **checkboxes** to select which assessment types to generate.

Step 2: Click **Generate Assessments**.

Step 3: The AI agent creates questions, expected answers, and K/A mappings for each type.

5.3 Downloading Assessments

- **Individual downloads** — Each assessment type has its own download button.
- **Bulk download** — Download all as assessments_[CourseName].zip

6. Generate Slides

Generates a presentation slide deck with infographics for the course content. The system uses a multi-agent pipeline to create content-rich slides with visuals.

6.1 Generating Slides

Step 1: Navigate to **Generate Slides** in the sidebar.

Step 2: Ensure course info has been extracted (Section 2).

Step 3: Click **Generate Slides** to start the multi-agent pipeline.

The pipeline runs through 5 phases:

- **Phase 1: Research** — Researches topic content
- **Phase 2: Content** — Generates slide content blocks
- **Phase 3: Editor** — Refines and structures content
- **Phase 4: Infographic** — Creates visual infographics
- **Phase 5: Assembly** — Builds the final PPTX file

6.2 Downloading the Slide Deck

Once complete, use the **Download** button to save the PPTX presentation file. The slide deck includes company branding, WSQ logos, and topic infographics.

Course Duration	Target Slide Count
1-day course	100 slides
2-day course	140 slides
3-day course	210 slides
4-day course	250 slides

7. Generate Brochure

Creates a marketing brochure PDF by scraping course information from a webpage.

7.1 Entering Course URL

Step 1: Navigate to **Generate Brochure** in the sidebar.

Step 2: Enter the **Course URL** — the link to the course page (e.g., from MySkillsFuture or the training provider's website).

Note: If you already entered the Course URL during Extract Course Info (Section 2), it will be auto-filled here.

7.2 Generating & Downloading

Step 1: Click **Generate Brochure**.

Step 2: The system scrapes the webpage for course details, fees, requirements, etc.

Step 3: A branded PDF is generated with company logo and WSQ branding.

Step 4: Click **Download** to save the PDF brochure.

8. Convert Documents

Convert existing (non-standard) documents into standardized WSQ format. This page has **three tabs**:

8.1 Convert Assessments

Step 1: Go to the **Convert Assessment** tab.

Step 2: Upload one or more assessment .docx files (question papers, answer keys).

Step 3: Click **Convert**. The AI extracts questions, answers, and K/A mappings.

Step 4: Download the converted files as a ZIP — each file is in standard WSQ format.

8.2 Convert Courseware

Step 1: Go to the **Convert Courseware** tab.

Step 2: Upload AP, FG, LG, or ASR documents (.docx).

Step 3: Click **Convert** to reformat into WSQ standard.

8.3 Convert Lesson Plans

Step 1: Go to the **Convert Lesson Plan** tab.

Step 2: Upload existing lesson plan .docx files.

Step 3: Click **Convert** to restructure into the standard timetable format.

9. Courseware Audit

The audit tool cross-checks your generated courseware documents against the Course Proposal (source of truth) to catch any mismatches or errors.

9.1 Quick Check Mode

If you have just generated documents on the AP/FG/LG page, you can use **Quick Check**:

Step 1: Navigate to **Courseware Audit** in the sidebar.

Step 2: Click **Quick Check** — it automatically audits the documents you just generated using the session data.

Step 3: Review the audit table: green checkmarks (✓) for matches, red crosses (✗) for mismatches.

9.2 Upload & Check Mode

For auditing existing documents (not just generated ones):

Step 1: Upload the **Course Proposal** (.docx, .xlsx, or .pdf).

Step 2: Upload the **courseware documents** to audit (AP, FG, LG, LP).

Step 3: Click **Audit**. The AI compares fields across all documents.

Step 4: Review the results table showing: field name, CP value, document value, and status.

Fields checked include:

- Course title, TSC code, TSC title
- Learning outcomes and topics
- Assessment methods and durations
- Organization name and UEN
- Proficiency levels and credit hours

10. Company Management (Settings)

Manage the training provider companies whose branding appears on generated documents.

10.1 Adding a Company

Step 1: Click the **Companies** button at the bottom of the sidebar.

Step 2: Click **Add New Company**.

Step 3: Fill in the details:

- **Company Name** — Full legal name
- **UEN** — Unique Entity Number (e.g., 20120096W)
- **Address** — Registered address
- **Email** — Contact email
- **Phone** — Contact number

Step 4: Upload the **company logo** (PNG or JPG). This appears on all documents.

Step 5: Click **Save**.

10.2 Editing / Deleting a Company

- **Edit:** Click the edit button next to any company to update its details or logo.
- **Delete:** Click delete — the system auto-backs up the company data before removing it (saved in `company/deleted_companies/`).
- **Search:** Use the search bar to find companies by name, UEN, or address.

11. Troubleshooting & FAQ

Problem	Solution
App won't start / port in use	Close all Streamlit processes: taskkill /F /IM streamlit.exe (Windows) Then restart: uv run streamlit run app.py
'No course info loaded' warning	Go to Extract Course Info (page 1) and upload a Course Proposal first. All other pages depend on this.
Generation is stuck / spinning forever	Check the Agent Status in the sidebar. If a job is stuck, refresh the page (F5) and try generating again.
Generated document has wrong company name	Check the Company dropdown in the sidebar — make sure the correct company is selected before generating.
Downloaded DOCX looks wrong / formatting issues	Open the file in Microsoft Word (not Google Docs or LibreOffice) for best results. WSQ templates are optimized for MS Word.
Brochure generation fails	Ensure the Course URL is valid and accessible. Some websites block automated scraping.
Slides have fewer than expected	The system auto-pads to hit the target count. If still short, try regenerating.
Need to re-extract course info	Simply upload the CP again on the Extract Course Info page. It will overwrite the previous extraction.

Quick Reference — Recommended Workflow

Step	Page	Action	Output
1	Extract Course Info	Upload CP + enter TGS code	Extracted JSON (session)
2	Generate AP/FG/LG	Select docs → Generate	AP, ASR, FG, LG (.docx)
3	Generate Lesson Plan	Click Generate	LP (.docx)
4	Generate Assessment	Select types → Generate	Assessment papers (.docx)
5	Generate Slides	Click Generate	Slide deck (.pptx)
6	Generate Brochure	Enter URL → Generate	Brochure (.pdf)
7	Courseware Audit	Quick Check or Upload	Audit report